Terms of reference (ToR) for Developing Integrated Frameworks and Digital Solutions for FLR Monitoring, Evaluation, and Reporting under RECAP4NDC project

Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH

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0. List of abbreviations

AFOLU – Agriculture, Forestry, and Other Land Uses

ARIES – Artificial Intelligence for Ecosystem Services

AVB – General Terms and Conditions of Contract (Allgemeine Vertragsbedingungen)

DC – Development Cooperation

ES – Ecosystem Services

FLR – Forest Landscape Restoration

FSI – Forest Survey of India

GIZ – Deutsche Gesellschaft für Internationale Zusammenarbeit

ICFRE – Indian Council for Forestry Research and Education

ICIMOD – International Centre for Integrated Mountain Development

IUCN – International Union for Conservation of Nature

IoT – Internet of Things

MER – Monitoring, Evaluation, and Reporting

MoEFCC – Ministry of Environment, Forest, and Climate Change (India)

NCR – National Capital Region (Delhi and surrounding areas)

NDC – Nationally Determined Contribution

SFD – State Forest Departments

SOP – Standard Operating Procedure

STAC – SpatioTemporal Asset Catalog

TAG – Technical Assessment Grid

TERI – The Energy and Resources Institute

ToF – Trees outside Forests

TOR – Terms of Reference

UN – United Nations

WP – Work Package

1. Context

Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), GmbH – India is implementing a project with the title "Restore, Conserve and Protect Forest and Tree Cover for NDC Implementation in India (RECAP4NDC)" under the Indo-German bilateral cooperation. The Ministry of Environment, Forest, and Climate Change (MoEFCC), Government of India is the Indian political partner. The project is implemented by a 6-member consortium (GIZ -overall coordination, International Union for the Conservation of Nature -IUCN- India office, Forest Survey of India -FSI, The Energy and Resources Institute -TERI, Indian Council for Forestry Research and Education -ICFRE and the International Centre for Integrated Mountain Development -ICIMOD). The project's intended impact is to support India achieve its targets on restoring degraded forests and landscapes inside and outside forests as defined in the National Forest Policy and the NDC thereby improving livelihoods of forest dependent communities and farmers. The project's approach is in line with the principles of the Bonn Challenge, which indicate that Forest Landscape Restoration (FLR) is the ongoing process of restoring the ecological functionality of degraded and deforested landscapes while enhancing the well-being of people who coexist with these places. It aims to strengthen the resilience of landscapes and keep future management options open. By 2029, RECAP4NDC aims at achieving ecological, socio-economic, governance, and climate change benefits in 0.4 million ha of the forest landscape. 10 million people are expected to be benefitted from improved forest Ecosystem Services (ES). Upscaling will lead to the restoration of 1.1 m ha and 110 mtCO2e in 10 years.

Under the RECAP4NDC project, four states have been selected for cooperation and briefly characterised by a range of specific starting conditions:

- **The Delhi National Capital Region (NCR)** comprising Delhi and the neighbouring districts from the states of Uttar Pradesh, Haryana, and Rajasthan demonstrates the challenges of rapid urban development and encroachment with resulting stresses on green spaces and forests, due to competing development needs on limited land. At the same time, the NCR presents the interesting potential to engage in the restoration of urban tree cover and to embed such efforts in cross-agency collaboration. FLR approaches also provide an important avenue to engage various neighbourhood groups and organisations in larger restoration campaigns and to foster environmental education and awareness raising.

- **Gujarat** comprises several habitats and Agro-climatic zones, such as grasslands, mangroves, and mudflats. It exposes specific challenges such as coastal degradation and pressures by grazing activities in grasslands, requiring specific approaches for landscape restoration and sustainable management.

- **Maharashtra** accommodates 9 Agro-climatic zones such as Central Plateau, the Eastern Vidarbha and Coastal Zones. The state government is aware of the different challenges of FLR in these different environments. The state provides a potential for the comparative testing of different approaches in up to five to six potential sites across various Agro-climatic zones.

- **Uttarakhand** with its topography along different – mostly hilly and mountainous – elevation zones has already implemented different forms of community forestry via the strengthening of village forest management committees (Van Panchayats), microplanning, fund management, spring-shed management, and promotion of non-timber forest products. There are opportunities to supplement these programmes with

targeted interventions around forest fire management, invasive species management and strengthening existing community forestry structures.

Project outputs

The project outputs are as follows. Each Output is led by one consortium partner with active contributions from other partners.

- **Output I - FLR model implementation:** Model approaches for forest landscape restoration including sustainable forest management, sustainable management of trees and vegetation outside forests and agroforestry, are implemented in selected states in cooperation with stakeholders from public service, agriculture, private sector, and advisory services.
- **Output II - Monitoring, evaluation and reporting:** Integrated systems for Monitoring, Evaluation and Reporting (MER) of ecological and socio-economic benefits of FLR are used.
- **Output III - Financing FLR:** Models and tools for financing FLR from private, public and international sources are applied.
- **Output IV - Policy and planning:** Policies and guidelines for forest landscape restoration are anchored in existing policy and planning processes.
- **Output V - Capacities, knowledge and communication:** Human capacities, knowledge and communication mechanisms for FLR have been transferred to and communicated among involved stakeholder groups.

Forest Landscape Restoration (FLR) initiatives are crucial for restoring degraded ecosystems, enhancing biodiversity, and improving socio-economic conditions in affected areas. Effective monitoring, evaluation, and reporting (MER) systems are essential to measure the success and impact of these initiatives. However, existing monitoring mechanisms in India, particularly in the Agriculture, Forestry, and Other Land Uses (AFOLU) sectors, face challenges related to data reliability, integration across governance levels, and technological adaptation.

The Ministry of Environment, Forest and Climate Change (MoEFCC), along with key partners such as the Forest Survey of India (FSI), State Forest Departments (SFDs), and international agencies like the International Union for Conservation of Nature (IUCN) and Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), recognizes the need for a standardized and technologically advanced MER framework for FLR. The proposed initiative seeks to develop integrated frameworks and digital solutions to bridge gaps in current monitoring systems, ensuring robust data collection, enhanced decision-making, and effective reporting on national and international commitments.

The assignment has following main objectives:

The objective of this Terms of Reference (TOR) package is to design, develop, and implement integrated MER frameworks and digital solutions for FLR monitoring. The initiative aims to:

- Create standardized and scalable MER frameworks, including protocols, measuring tools, guidelines, and evaluation methodologies to assess the ecological, economic, and social impacts of FLR interventions.
- Integrate innovative technologies such as remote sensing, drones, Internet of Things (IoT) devices, and artificial intelligence to enhance data collection, analysis, and visualization.

- Engage with SFDs, MoEFCC, FSI, and other key stakeholders through workshops and consultations to co-develop the MER framework, ensuring institutional buy-in and sustainability.
- Test the developed tools and methodologies in selected FLR sites to evaluate their effectiveness, feasibility, and adaptability to real-world conditions before full-scale implementation.
- Facilitate the integration of developed MER systems into national and subnational governance structures through capacity-building initiatives and institutional support mechanisms.

2. Tasks to be performed by the contractor

The assignment will be carried out to facilitate output II of the project, which focuses on integrated systems for Monitoring, Evaluation and Reporting (MER) of ecological and socio-economic benefits of FLR.

The qualified agency is responsible for providing the following services:

Work Package (WP) 1: Development of Integrated FLR Monitoring & Evaluation Systems

This work package focuses on designing comprehensive and adaptable Monitoring, Evaluation, and Reporting (MER) systems to assess the ecological, social, and economic impacts of Forest Landscape Restoration (FLR). It involves reviewing existing national and international monitoring frameworks, identifying key indicators, incorporating innovative digital technologies (such as remote sensing, drones and digital field data collection), and developing standard operating procedures (SOPs) and data collection formats. The goal is to create robust, scalable, and user-friendly MER frameworks that aid in decision-making and reporting at multiple governance levels. The MER framework will explore the possibility of incorporating drone data, supplied by GIZ and SFDs, to enhance the accuracy and efficiency of FLR monitoring and evaluation. The MER system should align with existing national (e.g., National Forest Inventory, India State of Forests Report) and international commitments (e.g., Bonn Challenge, UN Sustainable Development Goals, and the Paris Agreement).

The WP consists of, but is not limited to, the following activities:

Activity 1.1: Assess Existing Monitoring, Evaluation and Reporting Mechanisms, Processes and Frameworks in the AFOLU sectors relevant for FLR

<u>Tasks:</u>

- Conduct literature review and selective expert interviews on MER SOPs, tools, methodologies, mechanisms, processes in the fields of forestry, agriculture, and water.
- Conduct a literature review and stakeholder mapping (national, subnational, local) for organizations engaged in forest and biodiversity monitoring.
- 08 stakeholder meetings with different stakeholder groups while developing the concept and content.
- Specifically review the use of innovative and emerging digital technologies in FLR related fields in India and across the world.
- Review existing literature and methodologies for monitoring ecosystem services for FLR

- Identify the technological gaps and limitations in current practices, especially focusing on areas that could benefit from remote sensing, drones, IoT solutions and other technologies.
- Investigate the latest advancements in digital solutions and other technologies. that are applicable to FLR monitoring.
- Perform a feasibility study to evaluate the practicality and effectiveness of these technologies in the Indian FLR context.
- Determine the integration capabilities of new technologies with existing monitoring frameworks.
- Prepare an extensive list of relevant stakeholders in this field, including active public and private sector players in India and around the world, other bilateral and multilateral projects in India; to identify best practices
- Conduct interviews and organise stakeholder consultations (about 10, as necessary) to validate findings from the literature review and identify useful case studies for the project to learn from
- Explore monitoring mechanisms and processes at various levels of governance within the project's main partners, i.e. state forest departments, in consultation with consortium partners. Consider the current technical capacities, available technologies and human capacities of the forest departments in this exploration process.
- In coordination with the consortium, organise one-on-one discussions and stakeholder workshops to meet with other state forest departments to explore their MER mechanisms and processes.

Deliverables:

Baseline Report on Existing FLR Monitoring Mechanisms which includes,

- The findings from the review of current FLR monitoring practices, highlighting strengths, weaknesses, and areas for improvement.
- Existing monitoring and evaluation mechanisms, (sub-national, national and international);
- Findings from the assessment of current technologies and the feasibility study of potential technologies applications in FLR.
- Overview of the use of digital innovations in FLR and its potential uses for FLR in India, supported by stakeholder map highlighting key players in digital innovations for FLR
- Minutes of meetings/record of discussions from interviews and stakeholder consultations.

Activity 1.2: Recommend the design of MER frameworks for measuring FLR impacts

Tasks:

- Extensive discussions with the project's consortium partners, SFDs and the on-ground teams on the appropriate use of MER mechanisms for defining parameters and measuring FLR impacts.
- Specifically understand the capacities and needs of the stakeholders in the project's FLR sites.
- Define desired outputs and uses of monitoring systems in consultation with partners (SFDs, line departments, MoEFCC, FSI).
- Select set of proxy indicators based on literature review and on information from the sites selected within the 150.000 ha.

- Define monitoring parameters together with key stakeholders in the project sites. These can include key ecosystem services and socio-economic variables.
- Provide recommendations and feedback on the selection of digital tools, formats, indicators, methodologies for FLR monitoring and evaluation, augmenting/building on existing MER mechanisms and processes of the state forest departments and other line departments, as appropriate.
- Present specific options for MER mechanisms for the project's FLR sites which the project's consortium can select for piloting/roll-out.

Deliverables:

- Report on options and recommendations for MER mechanisms under the project, for the project's FLR sites, presented for consideration.

Detailed technical specifications for each developed digital monitoring tool, including hardware and software requirements, data capabilities, and user interfaces.

Work Package (WP) 2: Development of Digital Tools & Technologies for FLR Monitoring

This work package aims to design, prototype, and implement digital solutions that enhance FLR monitoring and reporting. It includes identifying technological gaps in existing monitoring mechanisms, developing user-friendly digital tools (such as mobile applications, dashboards, and remote sensing solutions), and integrating them with current national and subnational monitoring frameworks. The digital tools will be tailored to India's FLR context, ensuring seamless data collection, analysis, and reporting while facilitating institutional adoption through stakeholder training and capacity-building initiatives.

The WP consists of, but is not limited to, the following activities:

Activity 2.1: Design and implement user-friendly SOPs, data collection formats and tools (including digital tools) for efficient field data collection and analysis

Tasks:

- Review challenges and barriers identified during the FLR planning phase by IUCN in project states.
- Clarify with project stakeholders, specifically in the FLR sites, on the current monitoring SOPs, formats and tools used for data collection and monitoring (leading from Activity 1.2 in WP1).
- Collect baseline data on selected parameters/indicators in the project's pilot sites (ideally before or at the start of interventions).
- Match existing MER processes, mechanisms, SOPs, formats and tools with the requirements of the project's sites to see how the MER systems need to be augmented to monitor and report on ecosystem services arising from FLR implementation.
- Develop SOPs, formats and tools, including digital tools, for field data collection.
- Create user guides for project implementors and on-ground partners to use the developed MER SOPs, formats and tools
- Closely work with IUCN and on-ground partners to identify personnel who will be using the formats and tools.
- Support the on-ground teams of the project and the forest officials in modifying and adopting the new MER elements.

- Specifically on digital tools, consult with project stakeholders for suitable digital technology adoption for FLR and ToF monitoring. Consider factors like access to technology and language while designing the processes.
- Develop a digital solution plan for each project state/landscape/ecosystem.
- Present a package of solutions to stakeholders in each model site.
- Develop agreed-upon digital tools with relevant stakeholders, including discussions on technology stack and field testing.
- Support the procurement of digital tools and equipment.

Deliverables:

- SOPs for data collection.
- Data collection formats
- User guides and manuals for the augmented/newly developed MER elements
- Needs assessment for digital tools and digital tools solution plan, i.e. proof-of-concept/wireframe
- Technical specifications of digital tools clearly defined for procurement
- Develop agreed-upon digital tools with relevant stakeholders, including discussions on technology stack and field testing.

Activity 2.2: Establish Monitoring mechanisms at the institutional level for Implementation

Tasks:

- Identify relevant stakeholders for each tool based on factors like synergies, institutional objectives, and IT capacity.
- Review current IT infrastructure and assess training needs for shortlisted organizations.
- Develop cost estimates and plan IT support, team composition, and expertise required for tool implementation.

Deliverables:

- Report on technical specifications for each tool developed towards handover support.

Activity 2.3: Design and prototyping of digital monitoring tools

Tasks:

- Develop detailed specifications for new digital monitoring tools that utilise enhanced remote sensing, drones, IoT solutions, digitally collected field data and other technologies.
- Design tools that are capable of collecting, processing, and analyzing data on ecological, economic, and social impacts of FLR.
- Ensure that these tools are scalable, adaptable to different FLR environments, and user-friendly for stakeholders with varying levels of technical expertise.
- Build prototypes of the digital monitoring tools.
- Conduct iterative testing and refinement of these prototypes to ensure they meet the design specifications and project needs.

- Deploy the developed tools in selected pilot areas to test their functionality and impact in real-world settings.
- Monitor and evaluate the performance of these tools in collecting accurate and relevant FLR data.
- Integrate feedback from pilot testing to refine and finalise the tools.
- Develop comprehensive user manuals and training modules for the deployment and use of the new digital monitoring tools.
- Conduct 8 (eight) training sessions in hybrid mode for stakeholders to ensure effective adoption and utilisation of the technologies with support of FSI/GIZ.
- 08 regional workshops on the outcomes of assignment of each work package. The workshops will be organised by GIZ in collaboration with FSI and MoEF&CC.

Deliverables:

- Detailed technical specifications for each developed digital monitoring tool, including hardware and software requirements, data capabilities, and user interfaces.
- Reports documenting the development process, testing outcomes, and iterations of the digital monitoring tool prototypes.
- Report on analysis of the pilot testing phase, detailing the functionality, user feedback, and performance metrics of the new tools.
- Training manuals and user guides designed to assist stakeholders in effectively utilizing the new digital monitoring tools.

Certain milestones, as laid out in the table below, are to be achieved by certain dates during the contract term, and at particular locations:

Milestone/Deliverable	Deadline/place/person responsible
Baseline report (reviewed and approved) on existing Monitoring Mechanisms	3 Months after the start of the contract
Report on options and recommendations for MER mechanisms	6 Months after the start of the contract
Approved design and prototype of tool (for data collection and monitoring)	9 Months after the start of the contract
Development of FLR monitoring tools and SOPs (for data collection and monitoring)	18 Months after the start of the contract
Pilot testing of data collection and monitoring tools	22 Months after the start of the contract
Final report on pilot testing	30 Months after the start of the contract

Period of assignment: From 15th May 2025 until 31st December 2027

3. Data protection

The digital tool developed or upgraded on behalf of a local partner of GIZ must meet the highest data protection standards, especially those relating to data protection by design and by default, as stated in Annex 1. “Data protection standards for developing digital tools meant for GIZ's partners”. The contractor is therefore required to inform GIZ if any applicable national

requirement is incompatible with the provisions of this annex. We equally recommend the partner to conclude data protection agreements with the hosting service provider(s) and the maintenance service provider(s), where applicable. The GIZ would be available to support the partner whenever need arises.

4. IT/ Information security

- Network communication between the components of the application should be encrypted. Hard-coded keys (symmetric/asymmetric) should not be included in the application. If this is unavoidable, the handling of the key must be described. The information security risks related to the storage of the key must be evaluated. The transmission of authentication information (especially passwords) must be encrypted. The storage and transmission of sensitive and/or personal data must comply with current encryption standards. There must be a documented authorization concept for the application (e.g. in the operating manual).
- Security patches should be applied regularly (at least every 6 months) and event-related as soon as possible.
- A role and authorization concept must be implemented that includes at least the roles of system admin (full authorization to the entire system), CMS admin (+account management), editor and author (editing content/articles). The application to be developed must be operable with minimal system rights. Only the following user groups may have access to the backend: Developers and administrators of the Contractor as well as employees for editorial maintenance of the Contractor. Access to the backend at the operating system level may therefore only be granted to very limited user groups with the appropriate expertise. There must be a documented authorization concept for each application (e.g. in the operating manual). Changes to account data that are required for registration may only be made by the administrators and are to be blocked for the user.
- A password policy with complex password and protection against password stuffing must be implemented. Preferably, a two-factor authentication, especially for administrators, should be implemented and is to be possible via multiple channels (app, SMS or e-mail). Passwords must not be stored in plain text. Passwords may only be stored as hash values. The hash algorithm must conform to the current recommendations of the BSI (technical guidelines). Hard-coded passwords must not be included in the application. The transmission of authentication information (especially passwords) must be encrypted. If 2FA is compulsorily active for all users, it is possible to work with 8-digit passwords, for example, and to dispense with password expiration if necessary.
- Session cookies used for logins must be deleted after logging out from the client. In order to minimize operating errors (human errors), the ergonomics of the application should be designed safely according to the need for protection.
- The system must be hardened against SQL injection. In detail, this concerns the validation, filtering and cleansing of user input. The inputs may only have expected properties and characters and may not contain any unauthorized metacharacters that are passed to the SQL interpreter.
- The system must be protected against Cross Site Scripting (XXS) on the client side. The server(s) must be protected against reflected or persistent cross site scripting by securing the server source code. All data to be processed by the server must be validated before execution. Whitelists of permitted data can be used for this purpose. General conversion of certain script characters is also a popular method. It is to be prevented that executable metacharacters of the scripts are read by the server.

Cookies should only be read by the server (HttpOnly) and not by JavaScript in the browser.

- When implementing API interfaces, it is essential to harden them against malicious code injection (SQL injection, etc.) via the URL.
- For the upload of files, appropriate file filters must be provided so that no files with executable content (scripts, programs or SQL codes) can be uploaded and executed. Therefore additionally an up-to-date anti-virus software must be used.
- A system log is to be implemented in which at least logins and logouts of all users and actions such as updates, backups, uploads and downloads, changes to account data and authorizations, as well as all security-relevant actions and events are to be logged and documented. The inclusion of further log data is still to be discussed in detail with the project. The system log is to be documented in the operating manual.
- The error messages generated by the application (especially exception handling/exceptions) must not provide any information that allows conclusions to be drawn about the architecture or software/software versions used.
- The regular backups of the complete system are to be carried out by the contractor and checked accordingly for usability (restore). The backup can be stored on the server, but a copy must always be stored offline to prevent loss through hacker attacks. The intervals for the backups are to be coordinated with the project.
- Important backups always belong offline on another system and must be validated. A system log is to be implemented in which at least logins and logouts of all users and actions such as updates, backups, uploads and downloads, changes to account data and authorizations, as well as all security-relevant actions and events are to be logged and evaluated.
- The deletion procedure must be proven upon request. Legal retention obligations remain unaffected.
- Remote maintenance of systems may only be carried out via secure communication channels, such as HTTPS with an encryption standard of at least TLS 1.2 or higher. So-called VPNs can also be used for remote maintenance, these must be configured/secured at the endpoints by a firewall.
- An emergency manual should be prepared for the system with recovery and recovery plans.

5. Concept

In the bid, the bidder is required to show how the objectives defined in Chapter 2 are to be achieved, if applicable under consideration of further specific method-related requirements (technical-methodological concept). In addition, the bidder must describe the project management system for service provision.

Technical-methodological concept

Strategy: The bidder is required to consider the tasks to be performed with reference to the objectives of the services put out to tender (see Chapter 1). Following this, the bidder presents and justifies the strategy with which it intends to provide the services for which it is responsible (see Chapter 2). (Point 1.1.- TAG)

The bidder is required to present the actors relevant for the services for which it is responsible and describe the **cooperation** with them. (Point 1.2 – TAG)

The bidder is required to present and explain its approach to **steering** the measures with the project partners and its contribution to the results-based monitoring system.

The bidder is required to describe the key **processes** for the services for which it is responsible and create an **operational plan or** schedule (Point 1.4.1– TAG) that describes how the services according to Chapter 2 (Tasks to be performed by the contractor) are to be provided. In particular, the bidder is required to describe the necessary work steps and, if applicable, take account of the milestones and **contributions** of other actors in accordance with Chapter 2 (Tasks to be performed by the contractor) (Point 1.4.2– TAG).

The bidder is required to describe its contribution to knowledge management for the partner (Point 1.5.1– TAG) and GIZ and promote scaling-up effects (Point 1.5.2– TAG) (**learning and innovation**).

Project management of the contractor

The bidder is required to explain its approach for coordination with the GIZ project.

- The bidder is responsible for selecting, preparing, training, and steering the experts (international and national, short, and long term) assigned to perform the advisory tasks.
- The bidder makes available equipment and supplies (consumables) and assumes the associated operating and administrative costs.
- The bidder manages costs and expenditures, accounting processes, and invoicing in line with the requirements of GIZ.
- The bidder reports regularly to GIZ in accordance with the AVB of the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH from 2022.

The bidder is required to draw up a **personnel assignment plan** with explanatory notes that list all the experts proposed in the bid; the plan includes information on assignment dates (duration and expert days) and locations of the individual members of the team complete with the allocation of work steps as set out in the schedule.

The bidder is required to describe its backstopping concept. The following services are part of the standard backstopping package, which (like ancillary personnel costs) must be factored into the fee schedules of the staff listed in the bid in accordance with section 5.4 of the AVB:

- Service-delivery control
- Managing adaptations to changing conditions
- Ensuring the flow of information between GIZ and field staff
- Contractor's responsibility for seconded personnel
- Process-oriented technical-conceptual steering of the consultancy inputs
- Securing the administrative conclusion of the project
- Ensuring compliance with reporting requirements
- Providing specialist support for the on-site team by staff at company headquarters
- Sharing the lessons learned by the contractor and leveraging the value of lessons learned on site.

6. Criteria for Eligibility of firms

Bidders must submit the documentary evidence **(as per Annex A of Bidding conditions)** for the criteria's specified below. Also please refer document "*Grid for eligibility of consulting firms*" for more details.

1. Be a registered as national organization or entity.
2. In case of bidding consortia: Declaration by consortium

3. Average annual turnover for the last three financial years should be at least EUR 200,000.
4. The agency should have minimum 10 employees as on previous three years.
5. The agency must have handled at least 3 projects on involving the development of monitoring, evaluation, or reporting systems, environmental frameworks, or similar tasks, with a minimum commission value of EUR 10,000. Preference will be given to projects related to ecosystem restoration, forestry, or climate adaptation.
6. The agency must have 5 years of experience in developing digital tools in the areas of environmental monitoring, evaluation, or reporting.
7. The agency should have 3 years of experience in providing consulting services for state/national agencies and completion of tasks according to timelines.
8. The agency should have 2 years' experience in Agriculture, Forests and Other Land use (AFOLU) sectors.
9. The agency should have regional experience in India
10. The agency should have experience in Official Development Assistance (ODA).

7. Personnel concept

The bidder is required to provide personnel who are suited to filling the positions described, on the basis of their CVs (see Chapter 10), the range of tasks involved and the required qualifications.

The below specified qualifications represent the requirements to reach the maximum number of points.

Team Leader – Monitoring and Evaluation Specialist (2.1 – Technical Assessment Grid)

Tasks of the Team Leader

- Overall responsibility for delivering advisory packages, ensuring the quality of outputs and adherence to timelines.
- Coordinate and maintain effective communication with FSI, SFDs, GIZ, and other stakeholders involved in the project.
- Manage personnel, plan assignments, and oversee short-term experts within the project budget.
- Regularly report to FSI, SFDs, and GIZ, adhering to established deadlines and project requirements.
- Lead the technical design and validation of the FLR MER framework, including indicators, methodologies, and Standard Operating Procedures (SOPs).
- Provide technical guidance and quality control for all deliverables prepared by the project team.

Qualifications of the Team Leader

- **Education/training (2.1.1):** Master's degree in environmental sciences, forestry, natural resource management, or related fields.

- **Language (2.1.2):** Knowledge of English, level C1 in the Common European Framework of Reference for Languages (CEFR).
- **General professional experience (2.1.3):** At least 10 years of experience in designing and implementing Monitoring, Evaluation, and Reporting (MER) systems in the AFOLU (Agriculture, Forestry, and Other Land Use) sector.
- **Specific professional experience (2.1.4):** At least 5 years of experience developing frameworks or tools for ecosystem restoration, forestry, or climate adaptation.
- **Leadership/management experience (2.1.5):** At least 5 years of experience as a project manager or team leader.
- **Regional experience (2.1.6):** At least 4 years of experience working on projects in India.
- **Development Cooperation (DC) experience (2.1.7):** At least 2 years of experience working on projects funded by development cooperation organizations.

Expert 1 – Data scientist/ engineer (2.2 – Technical Assessment Grid)

Tasks of Expert 1

- Collect and analyze data on ecological, socio-economic, and governance indicators for FLR monitoring.
- Support the development of proxy indicators for ecosystem services and adaptation benefits, such as water security, slope stabilization, and biodiversity recovery.
- Prepare data collection templates and sampling designs aligned with project objectives.
- Design and develop data pipelines for FLR MER data processing and integration.
- Ensure interoperability between geospatial, remote sensing, and field-collected data.
- Develop and optimize databases for storing and managing large-scale FLR data.
- Implement cloud-based solutions for data storage, access, and analysis.
- Collaborate with web developers and data scientists to ensure seamless integration.
- Validate data accuracy and integrate findings into the FLR MER framework in collaboration with the FSI/SFD/GIZ project team.
- Provide technical input for pilot testing and refine methodologies based on field results.

Qualifications of Expert 1

- **Education/training (2.2.1):** Master’s degree in environmental sciences, forestry, natural resource management, or related fields.
- **Language (2.2.2):** Knowledge of English, level C1 in the Common European Framework of Reference for Languages (CEFR).

- **General professional experience (2.2.3):** At least 7 years of experience in data collection, analysis, and reporting for environmental or forestry-related projects.
- **Specific professional experience (2.2.4):** At least 4 years of experience using geospatial tools, remote sensing, or other technologies for environmental monitoring.
- **Regional experience (2.2.6):** At least 2 years of experience working on projects in India.

Expert 2 – Ecosystem Modeller (2.3 – Technical Assessment Grid)

Tasks of Expert 2

- Use ecosystem service models (e.g., InVEST, Natural Capital Accounting, ARIES) to assess ecological and socio-economic impacts of FLR interventions.
- Quantify ecosystem services such as carbon sequestration, water regulation, and biodiversity restoration.
- Develop scenarios to predict the outcomes of FLR initiatives under varying intervention strategies.
- Integrate ecosystem service valuation into the FLR MER framework.
- Conduct training sessions for stakeholders on the use of ecosystem models and their applications in FLR monitoring.
- Validate model outputs and align them with project objectives and stakeholder expectations.

Qualifications of Expert 2

- **Education/training (2.3.1):** Master's degree in ecology, environmental sciences, natural resource management, or related fields.
- **Language (2.3.2):** Knowledge of English, level C1 in the Common European Framework of Reference for Languages (CEFR).
- **General professional experience (2.3.3):** At least 7 years of experience in ecosystem modeling and service valuation.
- **Specific professional experience (2.3.4):** At least 4 years of experience applying tools such as InVEST, ARIES, or similar models for environmental assessment.
- **Regional experience (2.3.6):** At least 2 years of experience working on projects in India.

Expert 3 – Geospatial Expert (2.4 – Technical Assessment Grid)

Tasks of Expert 3

- Assess the applicability of technologies such as satellite imagery, GIS, AI, and machine learning for FLR monitoring.

- Develop geospatial methodologies for tracking ecological and socio-economic indicators, including vegetation cover, land-use change, and carbon stocks.
- Conduct geospatial analysis and produce high-quality maps and visualizations for reporting.
- Integrate geospatial tools into the FLR MER framework and prepare workflows for scalability across regions.
- Process and analyze satellite imagery to monitor FLR activities.
- Develop methodologies for tracking vegetation cover, land-use changes, and ecosystem health.
- Work with data engineers and scientists to integrate remote sensing products into digital tools.
- Validate remote sensing outputs through ground truthing and stakeholder consultations.
- Prepare training material and provide training sessions for stakeholders on the application of geospatial tools for monitoring and evaluation.
- Collaborate with the FSI/SFD/GIZ project team to validate proposed methodologies.

Qualifications of Expert 3

- **Education/training (2.4.1):** Master's degree in geospatial sciences, GIS, remote sensing, or environmental technology.
- **Language (2.4.2):** Knowledge of English, level C1 in the Common European Framework of Reference for Languages (CEFR).
- **General professional experience (2.4.3):** At least 7 years of experience in geospatial analysis, remote sensing, and related technologies.
- **Specific professional experience (2.4.4):** At least 4 years of experience in applying geospatial technologies for environmental monitoring and assessment.
- **Regional experience (2.4.6):** At least 2 years of experience working on projects in India.

Expert 4 – Software/ Web Developer (2.5 – Technical Assessment Grid)

Tasks of Expert 4

- Develop and maintain digital platforms for FLR MER data management.
- Design and optimize software and web applications/ dashboards for MER FLR.
- Ensure compatibility with GIS and remote sensing tools.
- Develop mobile and web applications for field data collection and real-time reporting.

- Ensure the software applications are user-friendly and compatible with GIS and remote sensing tools.
- Work with web developers and data engineers to optimize system performance.
- Conduct system testing, debugging, and performance tuning.
- Create technical documentation and provide training for end-users.
- Collaborate with stakeholders to integrate data visualization features.

Qualifications of Expert 4

- **Education/training (2.5.1):** bachelor's or master's degree in computer science, software engineering, or a related field.
- **Language (2.5.2):** Knowledge of English, level C1 in the Common European Framework of Reference for Languages (CEFR).
- **General professional experience (2.5.3):** At least 5 years of experience in web development.
- **Specific professional experience (2.5.4):** At least 3 years of experience in GIS-based application development.
- **Regional experience (2.5.6):** At least 2 years of experience working on projects in India.

8. Costing requirements

Assignment of personnel

Sr. No.	Key Personnel	No. of Days
Team leader	Team Leader – Monitoring and Evaluation Specialist	60
Expert 1	Data scientist/ engineer	120
Expert 2	Ecosystem Modeller	60
Expert 3	Geospatial Expert	250
Expert 4	Software/ Web Developer	250

Total proposed person days: 740

Travel

The agency is required to calculate the travel by the specified experts and the experts it has proposed based on the places of performance and list the expenses separately by daily allowance, accommodation expenses, and other travel expenses. Travel is anticipated to project sites and meeting locations in the following states:

- Maharashtra

- Gujarat
- Delhi NCR (mostly New Delhi, Haryana and Rajasthan)
- Uttarakhand

Sustainability aspects for travel

GIZ has undertaken an obligation to reduce greenhouse gas emissions (CO_2 emissions) caused by travel. When preparing your tender, please incorporate options for reducing emissions, such as selecting the lowest-emission booking class (economy) and using means of transport, airlines and flight routes with a higher CO_2 efficiency. For short distances, travel by train (second class) or e-mobility should be the preferred option.

CO_2 emissions caused by air travel must be offset. GIZ specifies a budget for this, through which the carbon offsets can be settled against evidence.

There are many different providers in the market for emissions certificates, and they have different climate impact ambitions. The Development and Climate Alliance (German only) has published a list of standards (German only). GIZ recommends using the standards.

A total of 90 domestic flights (45 outward and 45 return) between Delhi, India (or any other Indian city) and Ahmedabad, Nagpur and Dehradun of India take place as part of a consulting assignment, amounting to a total of EUR 1,198.00 (90 x EUR 13) is specified for the domestic flights, for bookings in economy class, category 1.3.

Fee days	**Number of experts**	**Number of days per expert**	**Total**	**Comments**
Team Leader	1	60	60	
National Experts	4	170	680	
Travel expenses	**Quantity**	**Number per expert**	**Total**	**Comments**
Per-diem allowance in country of assignment	80	5	400	for professional team (For workshops/ training and stakeholder meetings)
Transport	**Quantity**	**Number per expert**	**Total**	**Comments**
Domestic flights	18 (one way)	5	90	for professional team (For workshops/ training and stakeholder meetings)
CO_2 compensation for air travel	18 (one way)	5	90	A fixed budget of EUR **1,198** is earmarked for settling carbon offsets against evidence. (EUR 13 per economy flight – 10% offsetting costs for air travel within India)

Other costs

The agency should budget for the following additional expenses:

- Development of geospatial visualizations, graphics, maps, and analytical outputs to be included in project deliverables.

- Preparation of materials for stakeholder meetings, trainings and workshops, including digital templates, SOPs, and presentations.

- Software licenses (if applicable) for ecosystem modeling tools such as **InVEST** and **ARIES** or for geospatial platforms.

- Costs associated with documentation, printing, and report preparation.

9. Inputs of GIZ or other actors

GIZ and/ or other actors (e.g., consortium members) are expected to make the following available:

- Introduce the agency to the government administration at the start of the project.
- Lead all official communication with other actors for this assignment.
- Suggest additional participants for any meetings, trainings, and workshops.
- Any technical support required for the success of this initiative.
- Technical input for finalising the activity plan, review strategies and periodic review and reporting of progress.

10. Requirements on the format of the bid

The structure of the bid must correspond to the structure of the ToRs. In particular, the detailed structure of the concept (Chapter 3) is to be organised in accordance with the positively weighted criteria in the assessment grid (not with zero). It must be legible (font size 11 or larger) and clearly formulated. The bid is to be drawn up in English (language).

The CVs of the personnel proposed in accordance with Chapter 4 of the ToRs must be submitted using the format specified in the terms and conditions for application. The CVs shall not exceed 4 pages. The CVs must clearly show the position and job the proposed person held in the reference project and for how long. The CVs can also be submitted in English (language).

Please calculate your price bid based exactly on the aforementioned costing requirements. In the contract, the contractor has no claim to fully exhaust the days/travel/workshops/ budgets. The number of days/travel/workshops and the budget amount shall be agreed in the contract as ‘up to’ amounts. The specifications for pricing are defined in the price schedule.